

Project Report

Forge-X: Multi-Functional 4WD RC Rover with Single Robotic Arm

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1. Introduction

Remote-controlled mobile robots are becoming increasingly useful in areas such as **inspection, material handling, surveillance, hazardous-environment exploration, and education**. These robots allow users to perform tasks from a safe distance, especially in places that may be difficult or unsafe for humans to access.

Forge-X is a **4-wheel-drive (4WD) robotic rover** designed to provide both **movement and object manipulation**. Unlike a basic robot car that can only move from one place to another, **Forge-X** is equipped with a **robotic arm** that allows it to perform simple tasks such as **picking, lifting, and moving objects**.

The robot mainly consists of **three important subsystems**:

1. **Locomotion System:**

The four wheels provide 4WD movement, while the **skid-steer mechanism** allows the rover to move forward, backward, turn left and right, and maneuver effectively in confined spaces.

2. **Robotic Arm:**

A servo-controlled robotic arm is mounted on the rover. It provides basic manipulation capability, allowing the robot to **reach and handle lightweight objects**.

3. **Wireless Control System:**

The rover is controlled remotely using a **wireless RF communication link**. Commands from the operator are transmitted to the robot without requiring a physical connection.

All these subsystems are coordinated using an **ESP32 microcontroller**, which acts as the main control unit of the robot. It receives commands from the wireless controller and controls the motors and servo motors accordingly.

An **onboard LCD display** is also used to provide real-time information about the robot's current status, such as movement or control information. This gives the operator better awareness of what the robot is doing while it is being controlled.

Overall, **Forge-X integrates mobility, wireless communication, and robotic manipulation into a single platform**. This makes it a useful prototype for understanding concepts related to **embedded systems, robotics, motor control, wireless communication, and human-machine interaction**, while also providing a foundation for future improvements such as autonomous navigation, sensors, object detection, and more advanced manipulation.

2. Objectives

- Design and build a 4WD rover chassis capable of forward, backward, and in-place (zero-radius) turning.
- Implement a reliable skid-steer differential drive control scheme.
- Integrate a single robotic arm for basic object handling/interaction tasks.
- Establish a stable, low-latency 2.4 GHz RF wireless control link between operator and rover.
- Provide real-time operational feedback via an onboard display.
- Validate the system through practical testing of drive, turning, and arm operation.

3. Literature Survey

Skid-steer (differential) drive has long been favored in wheeled mobile robotics for its mechanical simplicity, since turning is achieved without a distinct steering mechanism, unlike Ackermann-steered vehicles — directly why Forge-X's zero-radius turning is possible without any dedicated steering linkage. Early work by Caracciolo, De Luca, and Iannitti established trajectory-tracking control for four-wheel differentially-driven robots, laying groundwork for the kind of forward/backward/turn control scheme used here. Mandow et al. later provided an experimental kinematic characterization of wheeled

skid-steer robots, showing how wheel/ground friction and slip affect real-world turning behavior — a practical concern for MDF-chassis builds like Forge-X operating on varied indoor surfaces. Kozlowski and Pazderski formalized a general kinematic and dynamic model for 4-wheel skid-steer robots, while more recent work by Rabiee and Biswas proposed friction-aware kinematic models that improve motion prediction accuracy for this drive type, and Martinez et al. extended skid-steer kinematic analysis to inclined terrain.

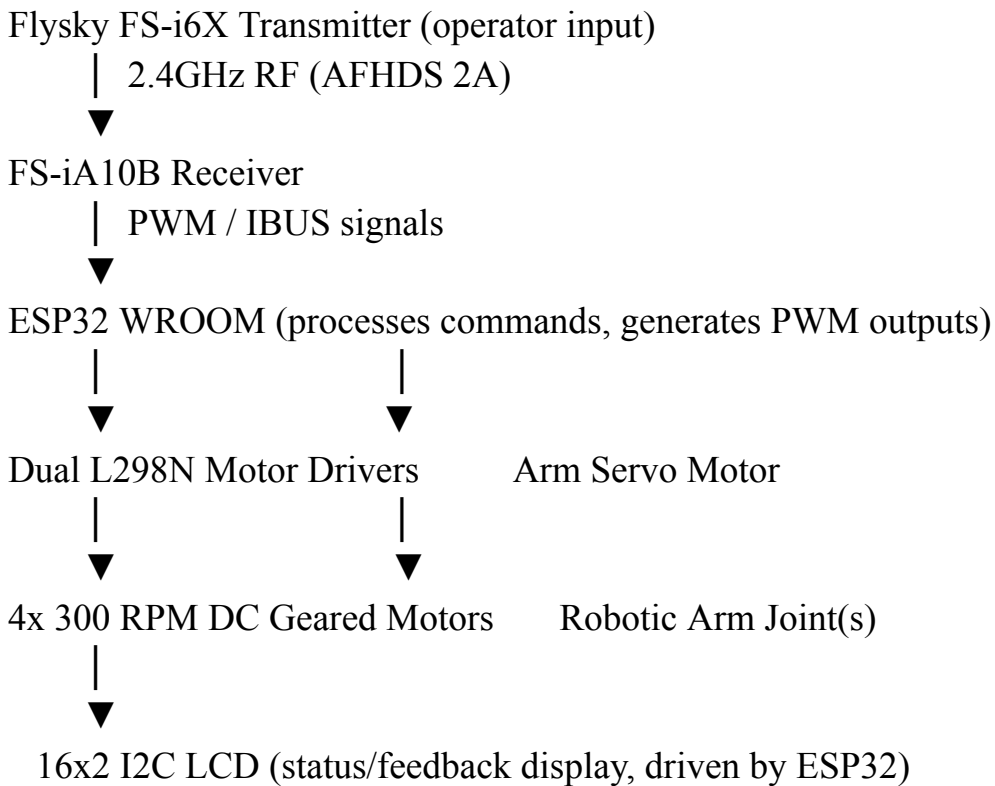
On the control and embedded-systems side, ESP32-based platforms have become a common choice for low-cost wireless robotics. Work on an ESP32-WROOM-32-based differential mobile robot for teaching robotics to undergraduate students mirrors Forge-X's own use of the ESP32 as a central controller for a differential-drive platform. Closer still to Forge-X's exact configuration, a 2024 IEEE conference paper describes a rover equipped with a robotic arm, remotely controlled through ESP32 microcontrollers, achieving seamless communication and control — directly comparable to the arm + drive + ESP32 integration used in this project, though that work adds object detection via YOLOv3, a possible future-scope direction. On the manipulation side, wireless-controlled robotic-arm designs using ESP32 and slider/servo-based control schemes have been documented as effective for basic angular positioning tasks, consistent with Forge-X's single-arm GPIO-PWM control approach.

4. System Overview

Forge-X integrates three functional subsystems:

Subsystem	Function
Locomotion	4WD skid-steer base for movement and turning
Manipulation	Single robotic arm with servo-driven joint(s)
Control& Communication	ESP32 + RF transmitter/receiver link, with LCD feedback

High-level signal flow:



5. Hardware Specifications

5.1 Chassis & Drive System

- **Frame material:** MDF (Medium Density Fiberboard) sheet
- **Drive configuration:** 4-wheel drive (4WD), skid-steer (differential) mechanism
- **Wheels:** 4x rubber-grip wheels
- **Motor mounting:** 4x metal L-brackets

5.2 Actuators

- **Wheel motors:** 4x DC geared motors, 300 RPM
- **Motor drivers:** 2x L298N dual H-bridge motor drivers (2 channels each, 4 channels total — one per wheel)
- **Arm actuator:** Servo motor for angular positioning (model/torque rating)

5.3 Power System

- **Battery type:** Lithium-Polymer (LiPo), high-discharge rated

- **Capacity:** 5200 mAh
- **Voltage:** (TBD — confirm cell count, e.g. 2S/7.4V or 3S/11.1V)
- **Logic power:** (TBD — likely a buck regulator stepping down to 5V for ESP32/LCD)
- Supplies: 4 DC motors, arm servo, ESP32 microcontroller, LCD display

5.4 Control & Communication

- **Microcontroller:** ESP32 WROOM (dual-core, multiple hardware PWM channels)
- **Transmitter:** Flysky FS-i6X, 2.4 GHz, 6-channel
- **Receiver:** FS-iA10B, 10-channel
- **Protocol:** AFHDS 2A (2.4 GHz RF)
- **Signal interface to ESP32:** PWM or IBUS

5.5 Feedback / Telemetry

- **Display:** 16x2 I2C LCD
- Shows real-time movement direction (e.g. "ACT: FORWARD/SPIN") and motor activity status

6. Working Principle

6.1 Control Architecture

1. The Flysky FS-i6X transmitter sends control signals over 2.4 GHz RF (AFHDS 2A protocol).
2. The FS-iA10B receiver picks up these signals.
3. The receiver forwards PWM/IBUS signals to the ESP32.
4. The ESP32 processes incoming channel data and generates corresponding PWM outputs.
5. Motor drivers and the arm servo execute the resulting commands.

6.2 Locomotion Mechanism (Skid-Steer Drive)

Motion	Left Wheels	Right Wheels
Forward	Forward	Forward
Backward	Reverse	Reverse
Turn Left (in place)	Backward	Forward
Turn Right (in place)	Forward	Backward

This differential logic enables **zero-radius turning** — the rover rotates about its own central axis, useful for navigating confined spaces without needing forward/backward clearance to turn.

6.3 Robotic Arm Operation

- The arm is controlled directly via ESP32 GPIO PWM output.
- The servo motor provides angular positioning of the arm joint.
- This enables basic object manipulation and controlled arm movement.
- (Degrees of freedom, payload capacity, and end-effector/gripper design — TBD)

6.4 Neutral/Stop Logic

The system treats the transmitter stick's neutral (50%) position as "Stop"; movement above or below this threshold maps to forward/backward or turning PWM outputs respectively.

7. Software Architecture

- **IDE:** Arduino IDE
- **Language:** C++ (clarify role of Python, if any — e.g. a ground-station tool)
- **Board configuration:** ESP32 by Espressif Systems, board: ESP32-WROOM-DA Module
- **Key libraries:** [LiquidCrystal_I2C.h](#), [Wire.h](#)

Control logic summary:

- Reads incoming RC channel values (via PWM/IBUS)
- Maps stick position to motor direction/speed per the skid-steer logic in §6.2
- Generates PWM signal to the arm servo based on a separate channel
- Updates the LCD with current action and motor status

8. Testing & Results

Parameter	Value
Top speed (PWM setting)	255
Battery life	30–45 minutes
Arm reach distance	TBD
Turning radius	Zero (in-place rotation)

Suggested additional tests:

- Load-carrying capacity of the arm
- Battery voltage drop under simultaneous drive + arm operation
- RF range/reliability test (max distance before signal loss)
- Turning accuracy/repeatability

Challenges faced:

- Voltage drop under combined motor + servo load, occasionally resetting the ESP32
- L298N heating and ~2V drop, reducing effective top speed
- Wheel scrub during skid-steer turns, requiring empirical PWM tuning
- Stick neutral drift causing creep; fixed with a software dead-band
- Chassis flex/misalignment causing the rover to veer off straight lines
- Servo jitter when drive motors ran simultaneously

9. Applications

- Educational demonstration of skid-steer robotics and RC-based control systems
- Confined-space object handling/retrieval
- Base platform for future sensor integration (obstacle avoidance, camera feed, etc.)
- Competition/exhibition robotics platform

10. Advantages & Limitations

Advantages:

- Simple, robust mechanical design (no steering linkage)
- Zero-radius turning ideal for tight spaces
- Reliable long-range wireless control via dedicated RC hardware
- Real-time operator feedback via LCD

Limitations:

- Skid-steer causes wheel scrubbing/friction losses and tire wear on hard surfaces
- Single arm limits simultaneous multi-object manipulation
- Battery life (30–45 min) constrains operational duration
- No autonomous capability (currently fully RC-operated)

11. Conclusion

Forge-X successfully demonstrates an integrated RC-controlled rover platform combining reliable 4WD skid-steer locomotion with single-arm manipulation, coordinated through an ESP32 microcontroller and a dedicated 2.4 GHz RF link. The system achieves zero-radius turning and real-time status feedback, making it well suited for confined-space navigation and basic object-handling tasks.

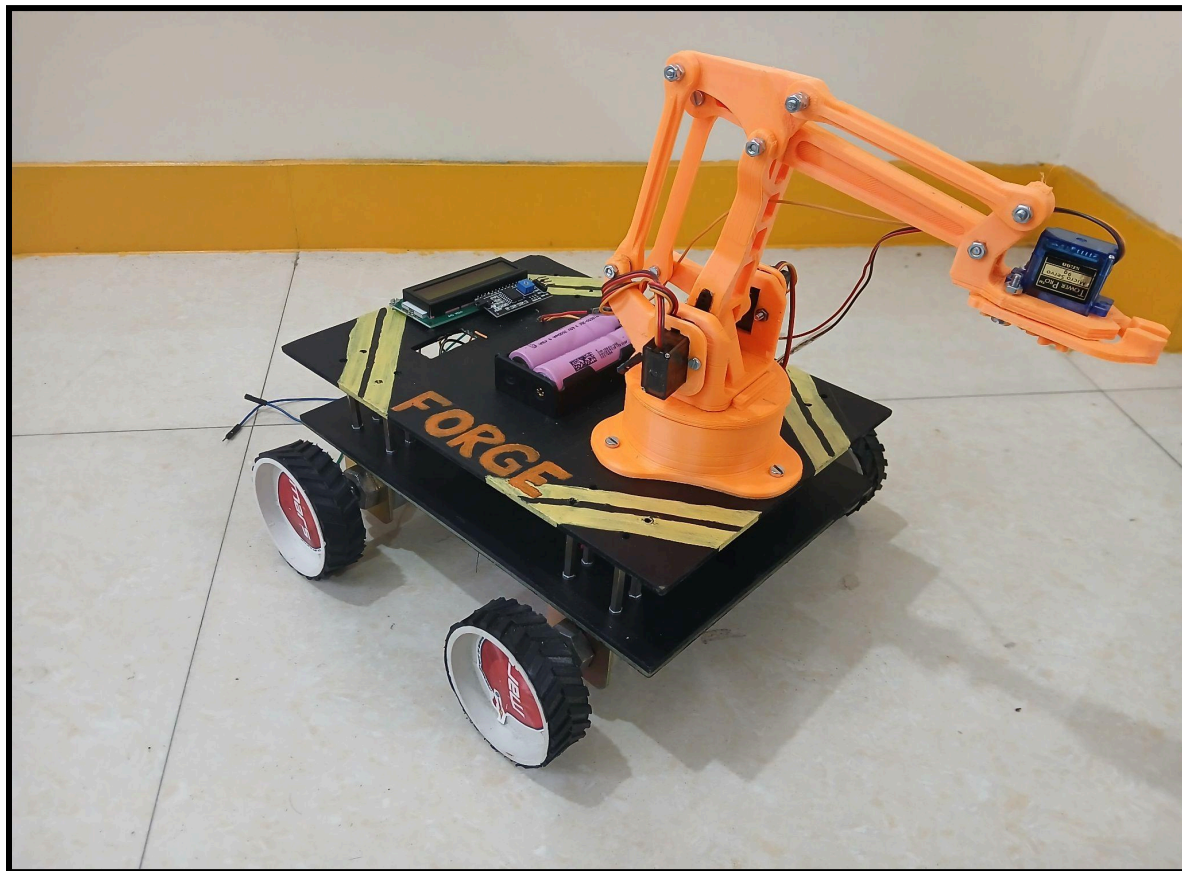
12. Future Scope

- Add autonomous navigation (ultrasonic/LIDAR-based obstacle avoidance)
- Camera feed for remote visual feedback (FPV)
- Closed-loop arm control (feedback from potentiometer/encoder for precision positioning)
- Battery management system (voltage/current monitoring displayed on LCD)
- Upgrade to a multi-DOF arm or add a gripper force sensor
- Object detection integration, as demonstrated in comparable ESP32-based rover-with-arm systems [7]

13. List of Materials

Component	Qty	Specification
MDF Chassis sheet	—	—
DC Geared Motor	4	300 RPM
L298N Motor Driver	2	Dual H-Bridge
ESP32 WROOM	1	Dual-core
Flysky FS-i6X Transmitter	1	6-channel, 2.4GHz
FS-iA10B Receiver	1	10-channel
16x2 I2C LCD	1	—
Servo motor (arm)	—	MG90
LiPo Battery	1	5200mAh
L-brackets	4	Metal
Wheels	4	Rubber-grip

14. Images



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